

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

What are data types?



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain different data types.
- I can explain how data is classified in different types.
- I can describe different types of data.

Ask your partner:

- How does a computer differentiate a number from a letter?
- Why is it important to separate different classes of data?



Let's Explore!

Discussing Data Types



Instructions: Read the text below.

- Johanna:** Hey, Martin. Do you know anything about data types? I don't understand that topic, and I am taking an exam next Monday. Do you have some time?
- Martin:** Actually, I am not doing anything right now. I can briefly explain to you what data types are. However, if you want to pass your exam, you must study hard this weekend. My explanation is not going to be enough.
- Johanna:** Yeah, I know. I have to pass this exam or I am failing the course. I am not going out this weekend; I am staying home and studying. So, what exactly are data types?
- Martin:** To begin, you must remember that data is just a series of bits in memory locations located in a circuitry of a computer. However, data types depend on the programming language that you, as a programmer, are using. Data types are very similar from language to language, but there are certain variations. The language normally specifies the kind of values for a determined data type. It also states how the computer processes and stores these values.
- Johanna:** That is very interesting. How about some examples of data types?
- Martin:** I can mention a few off the top of my head. Let's see. First, there are integers, which are whole numbers without fractional parts. Then, we have floating point numbers or floats, which are numbers that can contain a fractional part. Next, we find characters, which can be letters and symbols different from numbers. Finally, there are strings, which can be a combination of alphanumeric symbols. There are many others, but you have to study them by yourself.
- Johanna:** Thank you, Martin. You are very helpful. Sure, I am studying the whole weekend. I have to apply myself if I want to graduate.



Let's Practice!

Data: Classification, Types and Characteristics.

Instructions: Select the correct answer.

1. _____ integers the type of data information that uses a language with just numbers?
 - a. Are
 - b. Is
 - c. Does
 - d. Do
2. Are floats also called _____? floating point numbers
 - a. strings
 - b. characters
 - c. integers
3. Data is categorized based on some simple parameters. It _____ take into account the language variations.
 - a. must
 - b. must to
 - c. have
 - d. has



Let's Practice!

How is data classified?

Instructions: Fill in the gaps. Use: isn't classified, depend, I'm trying, Does

1. Data _____ at random. It follows some parameters that are compared in order to categorize the whole data information.
2. _____ data categorization _____ on the kind of values?
3. _____ to remember if language determines the type of data based on the kind of values.



Extension Activity

I can know more about data types

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about data types.
- Create a post with the main characteristics of data types.
- Share with your partners the post you created.

Self-Assessment

Yes No

- I look for more information about data types.
- I create a post with the main characteristics of data types.
- I share with your partners the post you created.

☐	☐
☐	☐
☐	☐



Target Words

- **alphanumeric** (adjective): composed of numbers and letters
e.g. Finally, there are strings, which can be a combination of alphanumeric symbols.
- **to apply oneself** (expression): to work or study hard
e.g. I have to apply myself if I want to graduate.
- **enough** (adjective): sufficient or adequate
e.g. My explanation is not going to be enough.
- **fractional** (adjective): division or part of a complete unit
e.g. Then we have floating point numbers or floats, which are numbers that can contain a fractional part.
- **state** (verb): to establish or determine
e.g. It also states how the computer processes and stores these values.
- **off the top of one's head** (expression): to remember without thinking much
e.g. I can mention a few off the top of my head.



Learning Points

Simple Present, Present Continuous and Be Going to in Negative Sentences and Questions

The **simple present** tense, the **present continuous** tense and the future tense with **be going** can all be used to express negative statements and ask questions in different contexts.

We normally use the present simple tense when we want to talk about routines or facts. In the negative form, the present simple has the following structure:

Subject + Do/Does + Not + Infinitive verb + Complement

For example:

*She **doesn't work** on C++.*

When asking questions, the simple present has the following structure:

Do/Does + Subject + Infinitive verb + Complement

For example:

***Do** you **know** how to recompile a JAVA code?*

On the other hand, we use the **present continuous** to describe activities happening right now or in a planned future, depending on the context and time complement. In the negative form, the **present continuous** has the following structure:

Subject + Verb to be + Not + Verb-ing + Complement

For example:

*We **aren't writing** any apps in C++ now.*

*He **isn't working** on that database for the next month.*

When asking questions, the **present continuous** has the following structure:

Verb to be + Subject + Verb-ing + Complement

For example:

***Are** you **determining** the kind of values?*

***Are** they **compiling** the floating point numbers?*





Learning Points

Finally, we use the future form **be going to** when we describe the intension to do something in the future. In the negative form, the future form **be going to** has the following structure:

Subject + Verb to be + Not + Going to + Infinitive verb + Complement

For example:

*They **aren't going to** include new features in this new version.*

When asking questions, the future form **be going to** has the following structure:

Verb to be + Subject + Going to + Infinitive verb + Complement

For example:

***Is he going to** include the missing characters?*

Examples

- *Do you know anything about data types?*
- *I don't understand that topic, and I am taking an exam next Monday.*
- *My explanation is not going to be enough.*
- *I am not doing anything right now.*
- *I am not going out this weekend.*

Modal Verbs Must and Have to in Affirmative Sentences

The modal verbs **have to** and **must** are used when we talk about obligations.

The modals **have to** and **must** express a strong obligation. However, each one has a slightly different application.

We use the modal **have to** in order to describe an obligation that does not depend on the speaker. In other words, the circumstances that make the situation obligatory are external to the person talking. The following are examples of the modal **have to** in affirmative sentences:





Learning Points

Examples:

Johan **has to** finish this code by 8 am.
They **have to** organize this database.

On the other hand, we use the modal **must** when talking about an obligation that relates to the personal circumstances of the person speaking. In other words, the speaker agrees with the obligation, which is in consonance with their beliefs. The following are examples of the modal must in affirmative sentences:

Examples:

I **must** hurry if I want to finish this code.
You **must** be organized when trying to recompile the whole code.

Examples

- I have to pass this exam or I am failing the course.
- I have to apply myself if I want to graduate.
- You must study hard this weekend.
- You must remember that data types depend on the programming language that you are using.